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Executive Summary

Senior AI Pipeline & Platform Engineer

Sovereign Data Solutions is seeking a highly skilled Senior AI Pipeline & Platform Engineer to join our team in Melbourne. This role involves designing, maintaining, and optimizing decoupled, local data orchestration pipelines written in Python, developing automated document parsing and data extraction frameworks using JSON and Markdown, and optimizing inference workloads for small language models and local compute infrastructure.

Key Responsibilities:

- Design and maintain decoupled, local data orchestration pipelines written in Python.
- Architect automated document parsing and data extraction frameworks using JSON and Markdown.
- Optimize inference workloads for small language models (SLMs) and local compute infrastructure.
- Build clean, print-ready document automation flows and analytical rendering systems.

Requirements:

- Strong competency with system optimization, Python script automation, structured data serialization (JSON/YAML).
- Experience setting up automated multi-stage background tasks and micro-pipelines.
- Familiarity with local model engineering tools (e.g., Ollama, local model deployment).
- Passion for pixel-perfect automated report generation or typesetting architectures.

About Us:

Sovereign Data Solutions is a highly specialized consultancy building sandboxed, air-gapped data platforms for defense, aerospace, and real estate enterprises. We prioritize robust, modular software design patterns over bloated API wrappers. Our team of experienced consultants and engineers are dedicated to delivering high-quality solutions that meet the unique needs of our clients.

History:

We have a rich history in technical consulting and project management. Our team has worked with various industries, including aerospace, defense, and real estate, and we pride ourselves on our ability to deliver innovative and effective solutions using advanced technologies.

Skills:

Our technical expertise is extensive, covering a range of skills and technologies. We are proficient in Python script automation, structured data serialization (JSON/YAML), system optimization, multi-stage background tasks and micro-pipelines, local model engineering tools (e.g., Ollama, local model deployment), and Apache Spark for big data processing. Our team is also skilled in using advanced machine learning frameworks such as PyTorch, TensorFlow, and scikit-learn for building AI pipelines.

Professional Experience

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*_Senior Technical Advisor & Industry Educator_* // Professional Associations & Advisory (2000 - Present)

Professional Associations & Advisory – Core Tasks & Skills

Core Tasks and Responsibilities
- Delivered specialized technical curriculum training for the CASBE Alliance under the Municipal Association of Victoria, instructing multi-disciplinary consulting groups and council officers on engineering-focused ESD tool metrics[cite: 1].
- Conducted professional webinars detailing complex statutory building codes, guiding planners and environmental design officers on auditing engineering submissions and requesting missing parameters[cite: 1].
- Instructed academic seminars and university guest lectures covering human thermal dynamics, physiological heat transfer indices, and microclimate stadium turf health configurations[cite: 3].
- Developed and deployed customized script repositories using Perl and Python over a multi-decade timeline to automate simulation pipelines, data ingestion frameworks, and advanced engineering physics calculators[cite: 3].
- Established and managed multi-node Linux high-performance cluster compute infrastructure with job-queuing engines to execute complex 3D ray-tracing and computational fluid dynamics analysis workloads[cite: 3].

Identified Technical Skills
- Professional Engineering Instruction & Technical Curriculum Delivery[cite: 1]
- Statutory Code Compliance Auditing & Regulatory Oversight[cite: 1]
- Scientific Computing & High-Performance Cluster Compute Architecture[cite: 3]
- Automated Workflow Scripting & Software Lifecycle Management (Python, Perl)[cite: 3]

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*_Director & Principal Systems Architect_* // Co-Perform Pty Ltd (2018 - Present)

Co-Perform Pty Ltd – Core Tasks & Skills

Core Tasks and Responsibilities
- Designed, built, and maintained a centralized, end-to-end backend cloud ecosystem and data architecture as the sole full-stack developer to manage multi-sector project registries, client assets, and building parameters.
- Engineered custom multi-repository Python API integrations linking low-code NAC databases, WorkflowMax, Pneumatic timesheets, and Google Drive storage systems.
- Architected a dedicated server-side "report builder" pipeline using PyLaTeX to automate data compilation, record consolidation, and high-fidelity PDF engineering reports across 10 permit and design compliance variants.
- Developed custom in-house UI/UX frontends and backends using Plotly Dash, served via Gunicorn and an Nginx reverse proxy network, to enable client-side simulation execution via an asynchronous backend queuing system.
- Configured, provisioned, and maintained local workstation and server infrastructure running Ubuntu 24.04 LTS, using Rclone for cloud file system mappings and ngrok for secure remote tunneling.
- Managed local DevOps automation by scripting custom systemd user services to ensure automated startup orchestration, crash recovery, and remote software deployment routines.
- Maintained a hardware-accelerated workstation containing an NVIDIA 5070-class GPU to handle local LLM and AI-assisted development workloads.
- Delivered statutory engineering assessments and structural compliance reporting (including NCC Section J building envelope auditing, thermal comfort profiles, and daylight/glare modeling) for tier-1 builders, developers, and councils including Kingston City Council, Multiplex, Hutchies, Nation Partners, and Architectus.

- Conducted advanced commercial modeling assignments, including 4-Star and 5-Star Green Star compliance certifications for infrastructure assets, 70-story high-rise tower NatHERS performance modeling (Melbourne Square Stage One), and historical building envelope thermal mitigations (132 Burnley Street).
- Developed technical built-environment training frameworks and webinars for the Municipal Association of Victoria (CASBE Alliance) and CASPI, training local council ESD officers and town planners on BESS tool utilization and Section J architectural report auditing.

Identified Technical Skills
- Full-Stack Software Engineering & Data Architecture (Python, Plotly Dash, PyLaTeX)
- Cloud Infrastructure, DevOps & Server Administration (Ubuntu, Nginx, Gunicorn, Systemd, Rclone)
- Hardware Acceleration & Local AI Engineering Workflows (NVIDIA GPU Computing)
- Statutory Building Compliance Auditing (NCC Section J, NatHERS, Green Star Ratings)
- Advanced Building Simulation Engineering (Thermal, Daylighting, Ventilation Analysis)
- Technical Curriculum Delivery & Professional Engineering Instruction

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*_Sustainable Systems Lead_* // Moreland Energy Foundation Ltd (MEFL) (2014 - 2018)

Moreland Energy Foundation Ltd (MEFL) – Core Tasks & Skills

Core Tasks and Responsibilities
- Programmed a predictive thermal and energy modeling tool utilizing custom code to optimize early-stage concept planning and right-size solar PV installations by mapping interval load profiles against prospective solar generation curves.
- Collaborated directly with DesignInc to evaluate public housing prototypes engineered for future climate scenarios, analyzing passive thermal comfort, ventilation effectiveness, and occupant heatwave exposure risks.
- Modeled building performance metrics including energy consumption, daylight penetration, and thermal comfort to support statutory town planning submissions, Sustainability Management Plans (SMPs), and public housing retrofit schemes.
- Conducted primary research and structured interviews with public housing tenants across Melbourne high-rise developments to document local thermal comfort challenges and formulate structural upgrade recommendations for the Department of Health and Human Services (DHHS).
- Orchestrated a regional insulation retrofit initiative for the Carlton public housing high-rise residential towers based on targeted asset performance evaluations.
- Engineered a large-scale hot water infrastructure modernization program across regional public housing, replacing obsolete gas facilities with efficient electric heat pumps and solar thermal systems.
- Programmed a centralized mobile data capture and site survey platform using Zoho Projects, enabling on-site deployment for field contractors capturing photographic and asset logs via tablet devices.
- Architected custom data-cleaning code pipelines to ingest, sanitize, and reconstruct highly fragmented and messy resident contact databases provided by public departments into operational data frameworks.
- Designed and executed comprehensive community net-zero transition analytics and energy modeling pipelines for the Hepburn Shire transition master plan using the Tableau visualization platform.

Identified Technical Skills
- Custom Energy Code Automation & Load Profiling
- Passive Thermal Design & Overheating Simulation
- Data Sanitation & Pipeline Refactoring (Custom Code)
- Mobile Infrastructure Engineering (Zoho Projects Architecture)
- Business Intelligence & Regional Energy Analytics (Tableau)
- Sustainable Infrastructure Program Management

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*_Building Physics Leader_* // Arup Australia (2005 - 2014)

Arup Australia – Core Tasks & Skills

Core Tasks and Responsibilities
- Served as Building Physics Leader coordinating multidisciplinary technical workflows across facades, mechanical, fire, acoustics, and sustainability engineering teams.
- Designed and built Linux high-performance computing (HPC) cluster architectures with queuing schedulers to manage advanced simulation workloads.
- Programmed custom data pipelines and analytical scripts using Perl to automate fluid dynamics, energy, and microclimate calculations.
- Developed and executed bespoke MATLAB bulk airflow models for a kilometer-high solar updraft tower (Enviromission), simulating mass flows, chimney updraft velocities, canopy skin friction, and ground thermal heat release.
- Conducted computational fluid dynamics (CFD) assessments for specialized urban applications, including structural wind drag predictions for the Southern Star Melbourne Wheel, fluid sloshing dynamics in storage tanks using ANSYS CFX, and jet fan car park ventilation profiles.
- Modeled detailed building envelopes using LBNL Window and Therm software to evaluate 2D facade framing components (mullions, transoms, edges), whole-system thermal U-values, and condensation risk.
- Performed regional stack emission and environmental plume dispersion simulations using TAPM and AUSPLUME within dense urban contexts, alongside rail tunnel ventilation, diesel pollutant, and train piston-effect studies.
- Conducted high-fidelity ray tracing for complex facade shading systems, solar reflectivity, and road/rail corridor glare risk assessments (including applications for Marina Bay Sands).
- Executed large-scale commercial HVAC energy models to demonstrate National Construction Code (NCC) Section J compliance for high-density architectures, such as the 10-story ANZ Docklands building incorporating absorption chillers and waterway heat rejection.
- Structured engineering compliance submissions for regional frameworks, including indoor air quality, daylight availability, and thermal comfort profiles under the Green Star Shopping Centre Design Pilot (Westfield Doncaster).
- Managed microclimate, canopy wind protection, and wind-driven rain analyses for international pedestrian infrastructure developments in Singapore.
- Delivered technical guest lectures on human physiological heat transfer, thermal comfort indices (PMV, PET, SET), and stadium turf health metrics (Photosynthetically Active Radiation) for university programs.

Identified Technical Skills
- Computational Fluid Dynamics (CFD) & High-Performance Computing Cluster Administration
- Building Envelope Thermal Analysis (LBNL Therm/Window)
- Custom Engineering Scripting & Modeling (MATLAB, Perl)
- Environmental Plume & Macro Pollutant Dispersion Modeling (TAPM, AUSPLUME)
- Whole-Building Energy Simulation & HVAC Performance Quantification
- Green Star & NCC Section J Statutory Compliance Auditing
- Solar Glare & Ray-Tracing Reflectivity Assessment

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*_Building Physics & Microclimate Specialist_* // Arup UK (2002 - 2005)

Arup UK – Core Tasks & Skills

Core Tasks and Responsibilities
- Executed computational fluid dynamics (CFD), radiant solar simulations, and thermodynamic modeling to

evaluate indoor airflow, building physics, and external microclimates.
- Modeled thermodynamic airflow profiles within complex multi-story atriums, building interior zones, and data centers.
- Conducted high-fidelity subgrid-scale turbulence CFD analyses of hot/cold aisle configurations and computer room air conditioning (CRAC) optimization arrays for data center infrastructure.
- Generated macro-scale environmental overlays and microclimate assessments for large-scale urban master planning initiatives, including a 17-hectare project in inner London with Rem Koolhaas.
- Conducted computational simulations to optimize outdoor and indoor occupant thermal comfort, pedestrian wind safety, facade daylight availability, and localized car exhaust pollutant dispersion.
- Performed fluid dynamics performance modeling for industrial gas turbine exhaust risers (such as Sakhalin in the North Pacific) to predict pressure drops, bypass modes, and heat-recovery efficiency parameters.
- Comparative evaluation of global climate zone parameters against local renewable options, thermal mass, and natural ventilation for a zero-carbon office buildings R&D framework.

Identified Technical Skills
- Computational Fluid Dynamics (CFD)
- Building Physics & Fluid Thermodynamics
- Urban Microclimate Assessment & Master Plan Modeling
- Data Center Airflow Optimization
- High-Fidelity Ray Tracing & Daylight Simulation
- Gas Turbine Heat-Recovery Modeling

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*_Computational Fluid Dynamics (CFD) & Wind Tunnel Specialist_* // British Maritime Technology (BMT) (2002 - 2005)

British Maritime Technology (BMT) – Core Tasks & Skills

Core Tasks and Responsibilities
- Executed computational fluid dynamics (CFD) explosion simulations using AutoReGas for offshore oil and gas platforms.
- Evaluated fluid turbulence around complex pipework layouts to assess blast overpressures and guide design modifications for blast containment walls.
- Modeled wave dynamics and interactions to quantify hydrodynamic impact loads on offshore substructures.
- Performed urban environmental modeling to analyze airflow around metropolitan buildings, evaluating pedestrian wind comfort profiles and urban pollutant dispersion.
- Prepared, configured, and executed wind tunnel testing programs using hardware scale models of municipal skyscrapers and industrial structures.
- Gathered high-resolution facade pressure distributions across building envelopes to determine structural design parameters for wind cladding loads.
- Conducted specialized aerodynamic testing and analysis for offshore facilities and rail rolling stock.

Identified Technical Skills
- Computational Fluid Dynamics (CFD)
- Wind Tunnel Testing & Scaled Model Instrumentation
- Industrial Explosion & Blast Overpressure Simulation (AutoReGas)
- Hydrodynamic Wave Load Quantification
- Urban Microclimate & Pollutant Dispersion Analysis
- Facade Cladding Load Analysis

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*_Consulting Engineer_* // Scott Wilson Irwin Johnston Pty Ltd (2000 - 2002)

Scott Wilson Irwin Johnston Pty Ltd – Core Tasks & Skills

Core Tasks and Responsibilities
- Conducted building energy simulation modeling to analyze and optimize building envelope performance.
- Undertook daylight and glare analysis for building developments.
- Modeled and evaluated natural ventilation strategies within the built environment.

Identified Technical Skills
- Sustainable Design
- Computational Modeling

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*_Research & Development Systems Engineer_* // Defence Science and Technology Organisation (DSTO) (1996 - 2000)

Defence Science and Technology Organisation (DSTO) – Core Tasks & Skills

Core Tasks and Responsibilities
- Performed computational fluid dynamics (CFD) assessments to analyze the trajectories of weapons released from F-111 aircraft to evaluate store clearance profiles.
- Executed simulations on Silicon Graphics workstations utilizing early turbulence models.
- Conducted acoustic and hydrodynamic analysis for the Collins class submarine focusing on cavitation surrounding the conning tower.
- Calculated lift and drag coefficients for the PC-9 trainer aircraft to generate data sets for flight simulators.
- Developed a digital twin simulation model for the F404 gas turbine engine using downhill simplex optimization methods to replicate engine test bed results and identify hardware component degradation factors.

Identified Technical Skills
- Computational Fluid Dynamics (CFD)
- Hydrodynamic and Acoustic Analysis
- Aerodynamic Coefficient Calculation
- Gas Turbine Performance Modeling & Digital Twins
- Optimization Methods (Downhill Simplex)